Hayden Farrell

[Email](#) | [linkedin](#) | [github](#) | [Website](#)

Education

University of Nottingham — Nottingham, UK
BSc Computer Science with a Year in Industry (Predicted First-Class Honours) — *Aug. 2023 – May 2027*

Aquinas College — Manchester, UK
Mathematics (A), Computer Science (A), Further Mathematics (A) — *Aug. 2021 – May 2023*

Experience

Software Engineer part time — Jul. 2026 – Present
Leonardo UK — *Newcastle upon Tyne, UK*

- One of only three undergraduate software engineers offered a part-time role while returning to university
- Developing safety-critical real-time software remotely as part of an Agile engineering team while completing my degree
- Balancing both university and work commitments to ensure deadlines are met
- Maintain and monitor Jenkins CI pipelines to ensure reliable build and test execution alongside ongoing software development.

Industrial Placement Software Engineer — June 2025 – July 2026
Leonardo UK — *Newcastle upon Tyne, UK*

- Developed and maintained safety-critical real-time software within an Agile engineering team following DO-178C development practices
- Led the design and implementation of a synthetic testing framework in C++ for complex embedded software, reducing dependency on hardware integration during development.
- Led design workshops with over 10 engineers including principal and lead software engineers to get stakeholder requirements and feedback on software design
- Developed a custom test application used by over 10 engineers across teams to allow for quick developer testing of a complex engineering system
- Created a custom tool to allow the visualisation of large amounts of realtime mathematical data to allow for quick analysis of complex engineering systems

Projects

Atlas | *C++, Vulkan, CMake, Python* — Current Project

- Developing a user-space heterogeneous task scheduler supporting CPU workloads and preemptive-style GPU scheduling through cooperative execution slices using Vulkan.
- Designed and implemented multiple scheduling policies, including FIFO, priority-based and round-robin, with benchmarking to evaluate latency, throughput and fairness
- Built a Vulkan compute backend enabling fine-grained control over GPU work submission, profiling execution using timeline visualisation and performance metrics
- Built an automated benchmarking framework to evaluate scheduling overhead and performance across different workloads, including CPU-bound, GPU-bound and mixed workloads

Blocky | *C++, OpenGL, GLSL, Python, CMake*

- Developed a voxel-based 3D game engine in C++ using OpenGL and GLSL
- Achieved a stable 60 FPS while rendering over 100,000 voxels with real-time lighting
- Created a custom linear algebra library to support vector and matrix mathematics required for 3D graphics

Technical Skills

Languages: C++, C, Python, Java, Matlab, VHDL, ARM, Rust, Shell Scripting, CMake
Libraries: OpenGL, Vulkan, Flask
Tools: Git, Docker, VS Code, Visual Studio, IntelliJ, Jenkins, MISRA, PolySpace
Training/Certificates: DO-178C Certified, C++ Training
Hobbies: Climbing, Bouldering, Leadership, Drones